S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 11

Aim: Develop a webpage demonstrating HTML5 APIs: Drag & Drop, Geolocation, and Web Storage.

Name of Student:

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AIM: Develop a webpage demonstrating HTML5 APIs: Drag & Drop, Geolocation, and Web Storage.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

To understand the role and integration of native HTML5 APIs in modern web applications.

To implement the Drag and Drop API for creating interactive and fluid user interfaces.

To utilize the Geolocation API to request and retrieve the geographical coordinates of a user's device.

To apply the Web Storage API (*localStorage* and *sessionStorage*) for efficient client-side data persistence without relying on server-side databases.

HARDWARE AND SOFTWARE REQUIREMENT:

Hardware Requirement

- 1.6 GHz or faster processor
- .RAM: 1GB
- Hard Disk Drive: > 80 GB

Software Requirement

- Operating System – Windows 2007,2010.
- Editor – Visual Studio Code /Sublime Text/Notepad.

THEORY:

1. **Introduction to HTML5 APIs**

   Application Programming Interfaces (APIs) in HTML5 provide developers with standardized JavaScript interfaces to interact with device hardware, browser features, and user interfaces natively, reducing the need for external plugins.

2. **Drag and Drop API**

   The HTML5 Drag and Drop API allows users to click and drag elements to a different location on the webpage.

   - *draggable="true"*: This attribute must be added to the HTML element you wish to make draggable.
   - Key Events:
     - *ondragstart*: Triggers when the user starts dragging an element.
     - *ondragover:* Triggers when a dragged element is over a valid drop target (default handling must be prevented using event.preventDefault() to allow dropping).
     - *ondrop*: Triggers when the dragged element is dropped onto the target.
   - *dataTransfer* Object: Used to hold the data that is being dragged during the drag-and-

drop operation.

3. **Geolocation API**

   The Geolocation API allows a web application to access the user's geographical location (latitude and longitude). Because this compromises user privacy, the browser will always prompt the user for permission before fetching the data.

   - Method: *navigator.geolocation.getCurrentPosition(successCallback, errorCallback)*
   - Use Cases: Mapping services, location-based weather updates, or finding nearby stores.

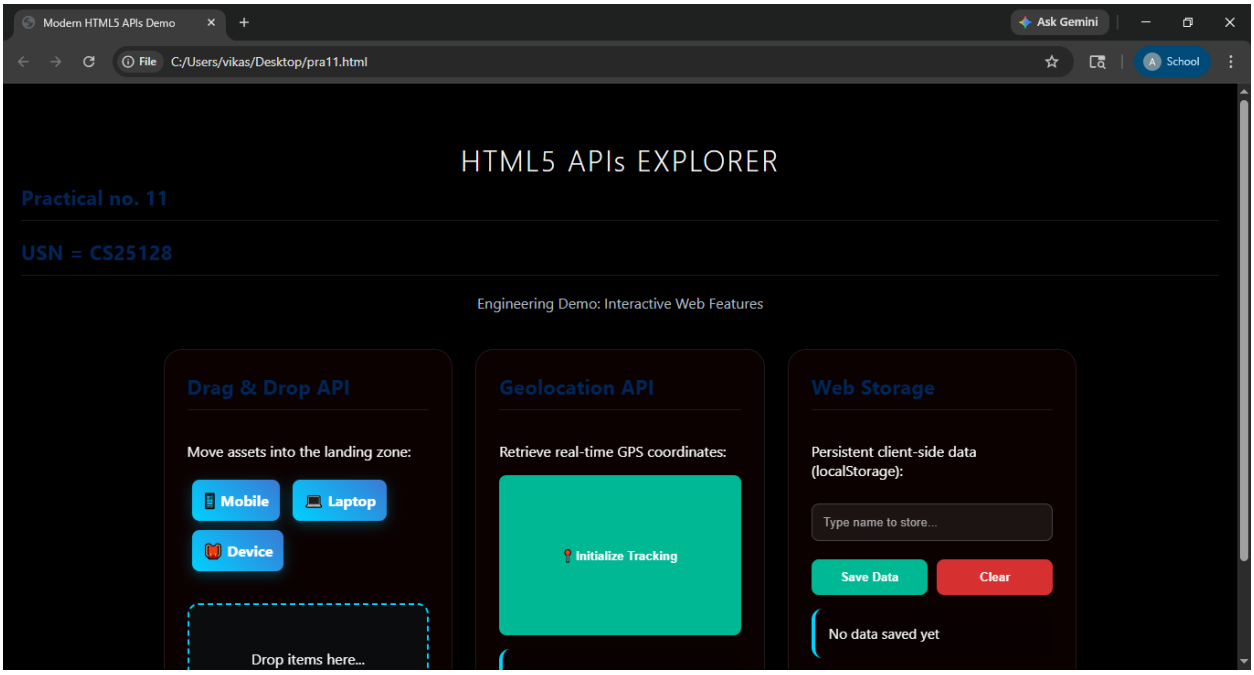
4. **Web Storage API**

   Before HTML5, application data had to be stored in cookies, included in every server request. Web Storage provides a more secure and larger capacity (usually around 5MB) mechanism to store key-value pairs locally in the browser.

   - *localStorage*: Stores data with no expiration date. The data persists even after the browser is closed and reopened.
   - *sessionStorage*: Stores data for one session. The data is deleted as soon as the browser tab is closed.
   - Common Methods:
     - *setItem(key, value)*: Saves data.
     - *getItem(key)*: Retrieves data.
     - *removeItem(key)*: Deletes a specific item.

CODE: (Write Code on loose practical pages)

OUTPUT/SCREENSHOTS: (Attach printout of output(s))



CONCLUSION:

Through this practical, I successfully implemented the **Drag & Drop**, **Geolocation**, and **Web Storage** HTML5 APIs to create a more interactive and functional web interface. I learned to manage user-driven events, retrieve real-time geographical coordinates with user permission, and store data efficiently on the client-side using localStorage and sessionStorage. Ultimately, this session demonstrated how native APIs reduce reliance on external plugins while enhancing the overall user experience.

DISCUSSION QUESTIONS:

1. What is the fundamental difference between localStorage and sessionStorage in the Web Storage API?
2. Why is it necessary to call event.preventDefault() inside the ondragover event handler?
3. How does Web Storage offer a better alternative to traditional HTTP Cookies for client-side data saving?
4. Explain why the Geolocation API requires explicit user permission before executing. What happens if the user denies the request?
5. Write a simple JavaScript snippet to save the username "Abhijeet" under the key "user" in localStorage, and then retrieve it.

REFERENCE:

- https://developer.mozilla.org/en-US/docs/Web/API/HTML_Drag_and_Drop_API
- https://developer.mozilla.org/en-US/docs/Web/API/Geolocation_API
- https://developer.mozilla.org/en-US/docs/Web/API/Web_Storage_API
- https://www.w3schools.com/html/html5_apis.asp